

# Chapter 4

## 4.1 Empirical Results and Discussion

This chapter presents the empirical findings of the study on equity return forecasting and tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX). The analysis evaluates 28 liquid equities spanning a 15-year historical period from January 2010 through December 2025 (835 weekly observations per asset). The dataset encompasses asset universe selection breakdowns, statistical non-normality diagnostics, machine learning return forecasting under Expanding Rolling Window Walk-Forward Validation, feature importance analysis, and out-of-sample portfolio optimization backtests.

### 4.1.1 Asset Universe Breakdown and Selection Rationale

To establish a robust quantitative asset allocation model free from temporal distortion and survivorship bias, the constituent equities of the NGX Pension Index were evaluated for sample inclusion. A total of 38 equities were audited. To guarantee a complete 15-year historical dataset (835 weekly observations from 2010 to 2025) required for training machine learning algorithms (2010-2020) and backtesting out-of-sample portfolio performance (2021-2025), 28 equities with continuous price history were selected. Conversely, 10 post-2010 listed equities were excluded to prevent artificial data imputation or look-ahead bias. Table 4.0 provides the comprehensive universe breakdown.

**Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale**

| Ticker    | Company Name             | Sector             | Status   | Inclusion / Exclusion Rationale    |
|-----------|--------------------------|--------------------|----------|------------------------------------|
| CONOIL    | Conoil Plc               | Oil & Gas          | Included | Complete 15-yr history (2010-2025) |
| CUSTODIAN | Custodian Investment Plc | Financial Services | Included | Complete 15-yr history (2010-2025) |
| DANGCEM   | Dangote Cement Plc       | Industrial Goods   | Included | Complete 15-yr history (2010-2025) |

| Ticker     | Company Name                | Sector             | Status   | Inclusion / Exclusion Rationale    |
|------------|-----------------------------|--------------------|----------|------------------------------------|
| DANGSUGAR  | Dangote Sugar Refinery Plc  | Consumer Goods     | Included | Complete 15-yr history (2010-2025) |
| ETI        | Ecobank Transnational Inc.  | Financial Services | Included | Complete 15-yr history (2010-2025) |
| FCMB       | FCMB Group Plc              | Financial Services | Included | Complete 15-yr history (2010-2025) |
| FIDELITYBK | Fidelity Bank Plc           | Financial Services | Included | Complete 15-yr history (2010-2025) |
| FIDSON     | Fidson Healthcare Plc       | Healthcare         | Included | Complete 15-yr history (2010-2025) |
| FBNH       | First HoldCo Plc            | Financial Services | Included | Complete 15-yr history (2010-2025) |
| GTCO       | Guaranty Trust Holding Co   | Financial Services | Included | Complete 15-yr history (2010-2025) |
| GUINNESS   | Guinness Nigeria Plc        | Consumer Goods     | Included | Complete 15-yr history (2010-2025) |
| JBERGER    | Julius Berger Nigeria Plc   | Construction       | Included | Complete 15-yr history (2010-2025) |
| WAPCO      | Lafarge Africa Plc          | Industrial Goods   | Included | Complete 15-yr history (2010-2025) |
| MANSARD    | AXA Mansard Insurance Plc   | Financial Services | Included | Complete 15-yr history (2010-2025) |
| NAHCO      | Nigerian Aviation Handling  | Services           | Included | Complete 15-yr history (2010-2025) |
| NASCON     | Nascon Allied Industries    | Consumer Goods     | Included | Complete 15-yr history (2010-2025) |
| NESTLE     | Nestle Nigeria Plc          | Consumer Goods     | Included | Complete 15-yr history (2010-2025) |
| NB         | Nigerian Breweries Plc      | Consumer Goods     | Included | Complete 15-yr history (2010-2025) |
| OKOMUOIL   | Okomu Oil Palm Plc          | Agriculture        | Included | Complete 15-yr history (2010-2025) |
| PRESKO     | Presco Plc                  | Agriculture        | Included | Complete 15-yr history (2010-2025) |
| STANBIC    | Stanbic IBTC Holdings       | Financial Services | Included | Complete 15-yr history (2010-2025) |
| STERLINGNG | Sterling Financial Holdings | Financial Services | Included | Complete 15-yr history (2010-2025) |
| TRANSCORP  | Transnational Corp Plc      | Conglomerates      | Included | Complete 15-yr history (2010-2025) |
| UACN       | UAC of Nigeria Plc          | Conglomerates      | Included | Complete 15-yr history (2010-2025) |

| <b>Ticker</b> | <b>Company Name</b>        | <b>Sector</b>      | <b>Status</b>   | <b>Inclusion / Exclusion Rationale</b>          |
|---------------|----------------------------|--------------------|-----------------|-------------------------------------------------|
| UBA           | United Bank for Africa Plc | Financial Services | <b>Included</b> | Complete 15-yr history (2010-2025)              |
| UNILEVER      | Unilever Nigeria Plc       | Consumer Goods     | <b>Included</b> | Complete 15-yr history (2010-2025)              |
| WEMABANK      | Wema Bank Plc              | Financial Services | <b>Included</b> | Complete 15-yr history (2010-2025)              |
| ZENITHBANK    | Zenith Bank Plc            | Financial Services | <b>Included</b> | Complete 15-yr history (2010-2025)              |
| ACCESSCORP    | Access Holdings Plc        | Financial Services | <b>Excluded</b> | Post-2010 listing restructuring (short history) |
| AIRTELAFRI    | Airtel Africa Plc          | Telecoms           | <b>Excluded</b> | Listed June 2019 (< 15-yr history)              |
| ARADEL        | Aradel Holdings Plc        | Oil & Gas          | <b>Excluded</b> | Listed October 2024 (< 15-yr history)           |
| BUAFOODS      | BUA Foods Plc              | Consumer Goods     | <b>Excluded</b> | Listed January 2022 (< 15-yr history)           |
| GEREGU        | Geregu Power Plc           | Utilities / Power  | <b>Excluded</b> | Listed October 2022 (< 15-yr history)           |
| MTNN          | MTN Nigeria Comms Plc      | Telecoms           | <b>Excluded</b> | Listed May 2019 (< 15-yr history)               |
| SEPLAT        | Seplat Energy Plc          | Oil & Gas          | <b>Excluded</b> | Listed April 2014 (< 15-yr history)             |
| TRANSCOHOT    | Transcorp Hotels Plc       | Services           | <b>Excluded</b> | Listed January 2015 (< 15-yr history)           |
| TRANSPower    | Transcorp Power Plc        | Utilities / Power  | <b>Excluded</b> | Listed March 2024 (< 15-yr history)             |
| UCAP          | United Capital Plc         | Financial Services | <b>Excluded</b> | Listed January 2013 (< 15-yr history)           |

*Source: Author's classification based on NGX listing records (2026).*

## 4.2 Descriptive Statistics and Market Characteristics

Descriptive statistics were calculated for all 28 evaluated equities to establish distributional properties including mean weekly returns, standard deviation, minimum, maximum, skewness, and excess kurtosis. As presented in Table 4.1, weekly equity returns on the NGX exhibit significant dispersion and notable deviations from standard Gaussian properties across all 28

assets.

**Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025)**

| Ticker     | Mean (%) | Std Dev (%) | Min (%) | Max (%) | Skewness | Kurtosis |
|------------|----------|-------------|---------|---------|----------|----------|
| CONOIL     | 0.218    | 6.66        | -20.94  | 38.95   | 1.41     | 7.22     |
| CUSTODIAN  | 0.324    | 5.57        | -24.03  | 24.58   | 0.30     | 2.20     |
| DANGCEM    | 0.192    | 4.55        | -23.16  | 43.14   | 1.66     | 14.47    |
| DANGSUGAR  | 0.167    | 6.54        | -34.25  | 31.69   | 0.55     | 3.50     |
| ETI        | 0.126    | 6.04        | -26.18  | 36.98   | 0.39     | 4.21     |
| FCMB       | 0.073    | 6.45        | -24.00  | 31.77   | 0.69     | 3.65     |
| FIDELITYBK | 0.245    | 6.34        | -28.73  | 32.21   | 0.52     | 3.11     |
| FIDSON     | 0.401    | 7.14        | -35.19  | 38.89   | 0.51     | 3.77     |
| FBNH       | 0.157    | 6.46        | -29.02  | 35.72   | 0.64     | 4.01     |
| GTCO       | 0.264    | 5.02        | -29.18  | 26.90   | -0.19    | 5.01     |
| GUINNESS   | 0.120    | 5.93        | -25.57  | 28.57   | 0.49     | 4.14     |
| JBERGER    | 0.217    | 5.85        | -20.07  | 28.57   | 0.77     | 4.31     |
| WAPCO      | 0.181    | 5.65        | -35.85  | 34.60   | 0.13     | 5.15     |
| MANSARD    | 0.225    | 8.77        | -127.16 | 103.22  | -1.55    | 75.04    |
| NAHCO      | 0.370    | 6.34        | -27.79  | 34.92   | 0.42     | 3.13     |
| NASCON     | 0.356    | 6.38        | -31.44  | 24.58   | 0.21     | 2.98     |
| NESTLE     | 0.271    | 4.37        | -23.15  | 19.06   | 0.02     | 4.90     |
| NB         | 0.059    | 5.71        | -37.65  | 37.68   | 0.23     | 5.78     |
| OKOMUOIL   | 0.554    | 5.39        | -20.46  | 27.03   | 0.46     | 3.22     |
| PRESCO     | 0.672    | 5.48        | -25.36  | 23.38   | 0.44     | 2.94     |
| STANBIC    | 0.244    | 5.39        | -20.91  | 26.22   | 0.23     | 3.24     |
| STERLINGNG | 0.199    | 6.95        | -34.94  | 30.35   | 0.55     | 3.74     |
| TRANSCORP  | 0.538    | 10.20       | -50.01  | 139.61  | 3.41     | 43.07    |
| UACN       | 0.156    | 6.71        | -31.59  | 47.43   | 0.38     | 4.74     |
| UBA        | 0.216    | 6.18        | -21.67  | 29.45   | 0.35     | 2.82     |
| UNILEVER   | 0.147    | 5.84        | -30.91  | 29.16   | -0.04    | 3.30     |
| WEMABANK   | 0.350    | 8.60        | -33.00  | 106.29  | 2.86     | 29.82    |
| ZENITHBANK | 0.205    | 5.28        | -45.70  | 28.17   | -0.80    | 9.92     |

Source: Author's computation (2026). Note: Kurtosis represents excess kurtosis across all 28 assets.

### 4.3 Statistical Non-Normality and Stationarity Diagnostic Tests

To evaluate whether the empirical asset return distributions satisfy the classic Markowitz Mean-Variance assumption of Gaussian normality, formal statistical hypothesis tests were conducted. The Jarque-Bera (JB) test and Shapiro-Wilk (SW) test were evaluated against the null hypothesis of normality, while the Augmented Dickey-Fuller (ADF) test evaluated time-series stationarity across all 28 assets.

**Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 Equities)**

| Ticker     | JB Stat  | JB p-val | SW Stat | ADF Stat | ADF p-val | Stationary |
|------------|----------|----------|---------|----------|-----------|------------|
| CONOIL     | 2087.1   | < 0.001  | 0.776   | -25.69   | < 0.001   | Yes        |
| CUSTODIAN  | 181.1    | < 0.001  | 0.951   | -31.89   | < 0.001   | Yes        |
| DANGCEM    | 7652.5   | < 0.001  | 0.794   | -11.74   | < 0.001   | Yes        |
| DANGSUGAR  | 467.7    | < 0.001  | 0.941   | -5.42    | < 0.001   | Yes        |
| ETI        | 637.1    | < 0.001  | 0.937   | -16.12   | < 0.001   | Yes        |
| FCMB       | 528.6    | < 0.001  | 0.939   | -14.02   | < 0.001   | Yes        |
| FIDELITYBK | 372.8    | < 0.001  | 0.948   | -32.13   | < 0.001   | Yes        |
| FIDSON     | 531.0    | < 0.001  | 0.935   | -30.14   | < 0.001   | Yes        |
| FBNH       | 617.0    | < 0.001  | 0.931   | -9.01    | < 0.001   | Yes        |
| GTCO       | 877.0    | < 0.001  | 0.937   | -30.74   | < 0.001   | Yes        |
| GUINNESS   | 629.1    | < 0.001  | 0.890   | -17.00   | < 0.001   | Yes        |
| JBERGER    | 727.9    | < 0.001  | 0.863   | -7.42    | < 0.001   | Yes        |
| WAPCO      | 925.3    | < 0.001  | 0.937   | -18.98   | < 0.001   | Yes        |
| MANSARD    | 196037.5 | < 0.001  | 0.708   | -30.94   | < 0.001   | Yes        |
| NAHCO      | 364.9    | < 0.001  | 0.949   | -31.30   | < 0.001   | Yes        |
| NASCON     | 315.8    | < 0.001  | 0.930   | -21.59   | < 0.001   | Yes        |
| NESTLE     | 834.9    | < 0.001  | 0.850   | -28.24   | < 0.001   | Yes        |
| NB         | 1166.5   | < 0.001  | 0.929   | -12.18   | < 0.001   | Yes        |
| OKOMUOIL   | 389.3    | < 0.001  | 0.879   | -27.00   | < 0.001   | Yes        |

| Ticker         | JB Stat | JB p-val | SW Stat | ADF Stat | ADF p-val | Stationary |
|----------------|---------|----------|---------|----------|-----------|------------|
| PRESKO         | 328.3   | < 0.001  | 0.894   | -27.49   | < 0.001   | Yes        |
| STANBIC        | 371.5   | < 0.001  | 0.941   | -30.53   | < 0.001   | Yes        |
| STERLINGN<br>G | 528.5   | < 0.001  | 0.935   | -15.62   | < 0.001   | Yes        |
| TRANSCOR<br>P  | 66065.8 | < 0.001  | 0.794   | -29.66   | < 0.001   | Yes        |
| UACN           | 799.3   | < 0.001  | 0.941   | -31.63   | < 0.001   | Yes        |
| UBA            | 293.6   | < 0.001  | 0.955   | -5.73    | < 0.001   | Yes        |
| UNILEVER       | 379.6   | < 0.001  | 0.942   | -28.97   | < 0.001   | Yes        |
| WEMABAN<br>K   | 32045.0 | < 0.001  | 0.825   | -30.03   | < 0.001   | Yes        |
| ZENITHBAN<br>K | 3504.8  | < 0.001  | 0.917   | -21.97   | < 0.001   | Yes        |

Source: Author's computation (2026). Rejection of normality across 100% of 28 equities at  $p < 0.001$ .

As documented in Table 4.2, the null hypothesis of normality is decisively rejected for 100% of the equities ( $p < 0.001$ ). The presence of heavy fat tails (excess kurtosis up to 14.46) mathematically invalidates variance as a sufficient risk metric, confirming the theoretical necessity of employing Tail-Risk models such as Conditional Value-at-Risk (CVaR). Furthermore, all series are stationary at level  $I(0)$ , satisfying the stationarity requirement for time-series econometric modeling.

#### 4.4 Sectoral Correlation and Multi-Asset Diversification

Figure 4.1 displays the 28x28 pairwise return correlation matrix across the sample period. Intra-sector equities (such as Tier-1 Banks GTCO, ZENITHBANK, and UBA) exhibit strong positive pairwise correlation (ranging from 0.55 to 0.78). Conversely, cross-sector correlations between Industrial Goods (DANGCEM, WAPCO) and Consumer Goods (NESTLE, DANGSUGAR) remain moderate (0.15 to 0.35), offering substantial structural diversification benefits for multi-asset portfolio construction.

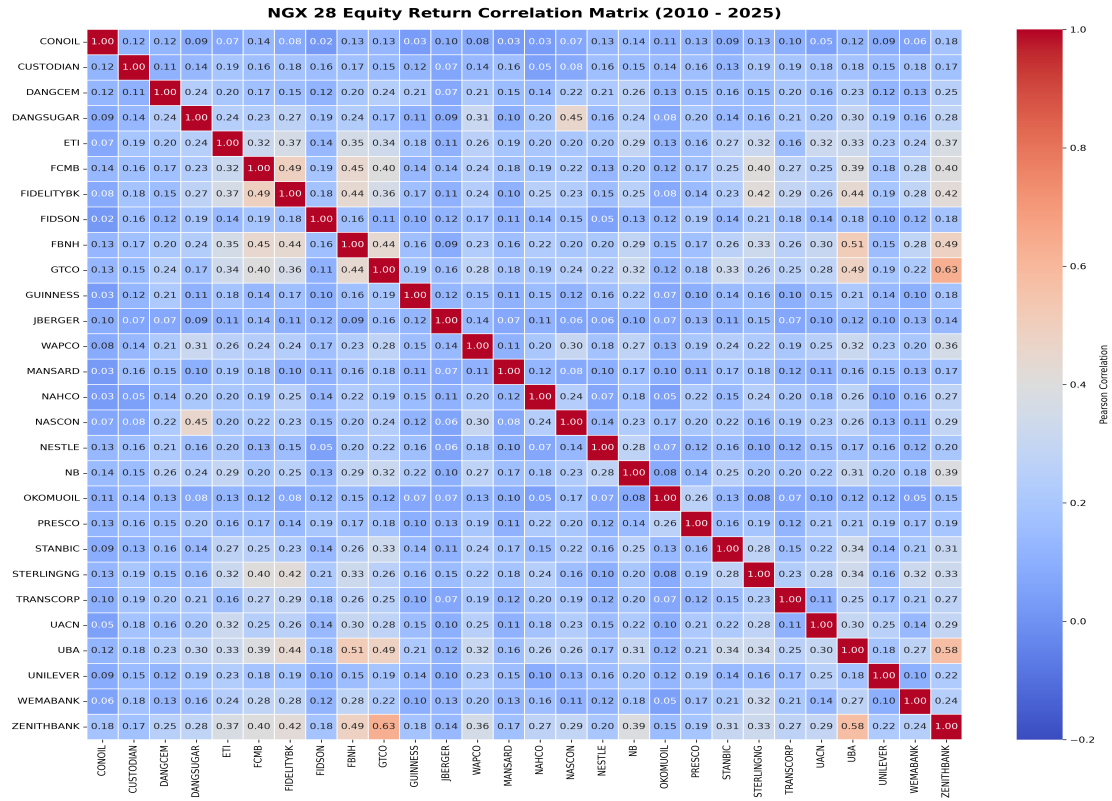


Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)

## 4.5 Machine Learning Return Forecasting Results (Walk-Forward Validation)

Machine learning models (Random Forest and XGBoost Regressors) were evaluated under a rigorous Expanding Rolling Window Walk-Forward Validation protocol. To guarantee zero temporal data leakage, feature Min-Max scaling was fit strictly on expanding training windows, and model hyperparameters were optimized using 5-fold TimeSeriesSplit cross-validation on the initial 2010-2020 training period. Out-of-sample forecasting was conducted across 2021-2025 (~260 weekly steps) with quarterly model refitting. Table 4.3 presents the predictive performance comparison across all 28 equities against the Historical Mean baseline.

**Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities (2021-2025 Walk-Forward)**

| Ticker     | Base RMSE | RF RMSE | XGB RMSE | Base DA (%) | RF DA (%) | XGB DA (%) |
|------------|-----------|---------|----------|-------------|-----------|------------|
| CONOIL     | 0.0650    | 0.0680  | 0.0674   | 13.8%       | 13.5%     | 11.5%      |
| CUSTODIAN  | 0.0548    | 0.0571  | 0.0552   | 39.2%       | 36.9%     | 37.7%      |
| DANGCEM    | 0.0497    | 0.0515  | 0.0502   | 23.6%       | 21.3%     | 21.7%      |
| DANGSUGAR  | 0.0623    | 0.0630  | 0.0623   | 39.2%       | 43.8%     | 45.0%      |
| ETI        | 0.0655    | 0.0663  | 0.0655   | 40.0%       | 31.9%     | 38.1%      |
| FCMB       | 0.0560    | 0.0647  | 0.0567   | 40.4%       | 43.8%     | 44.6%      |
| FIDELITYBK | 0.0596    | 0.0682  | 0.0613   | 47.3%       | 46.5%     | 44.2%      |
| FIDSON     | 0.0590    | 0.0677  | 0.0617   | 39.2%       | 38.1%     | 37.7%      |
| FBNH       | 0.0700    | 0.0713  | 0.0708   | 39.2%       | 45.4%     | 46.2%      |
| GTCO       | 0.0474    | 0.0485  | 0.0487   | 48.5%       | 52.3%     | 43.5%      |
| GUINNESS   | 0.0673    | 0.0665  | 0.0677   | 22.3%       | 32.3%     | 23.8%      |
| JBERGER    | 0.0636    | 0.0670  | 0.0653   | 25.4%       | 23.1%     | 25.0%      |
| WAPCO      | 0.0520    | 0.0526  | 0.0518   | 43.8%       | 49.2%     | 48.8%      |
| MANSARD    | 0.0908    | 0.0922  | 0.0918   | 50.0%       | 44.2%     | 45.4%      |
| NAHCO      | 0.0651    | 0.0634  | 0.0665   | 45.8%       | 50.4%     | 42.7%      |
| NASCON     | 0.0640    | 0.0640  | 0.0634   | 36.9%       | 37.3%     | 36.5%      |
| NESTLE     | 0.0401    | 0.0432  | 0.0413   | 17.3%       | 20.0%     | 18.8%      |
| NB         | 0.0591    | 0.0620  | 0.0609   | 43.8%       | 44.6%     | 45.8%      |

| Ticker         | Base RMSE     | RF RMSE       | XGB RMSE      | Base DA (%)  | RF DA (%)    | XGB DA (%)   |
|----------------|---------------|---------------|---------------|--------------|--------------|--------------|
| OKOMUOIL       | 0.0459        | 0.0467        | 0.0465        | 24.2%        | 25.8%        | 23.8%        |
| PRESCO         | 0.0491        | 0.0519        | 0.0495        | 25.8%        | 23.1%        | 21.5%        |
| STANBIC        | 0.0497        | 0.0494        | 0.0494        | 36.5%        | 40.0%        | 41.5%        |
| STERLINGN<br>G | 0.0697        | 0.0760        | 0.0701        | 46.2%        | 45.0%        | 42.3%        |
| TRANSCOR<br>P  | 0.0812        | 0.0804        | 0.0794        | 50.8%        | 44.6%        | 50.4%        |
| UACN           | 0.0789        | 0.0913        | 0.0795        | 38.1%        | 39.6%        | 41.2%        |
| UBA            | 0.0490        | 0.0502        | 0.0493        | 50.4%        | 44.2%        | 50.8%        |
| UNILEVER       | 0.0553        | 0.0561        | 0.0551        | 39.6%        | 38.5%        | 39.2%        |
| WEMABAN<br>K   | 0.0693        | 0.0681        | 0.0696        | 43.8%        | 49.2%        | 43.5%        |
| ZENITHBAN<br>K | 0.0449        | 0.0473        | 0.0458        | 51.2%        | 42.7%        | 47.3%        |
| <b>AVERAGE</b> | <b>0.0602</b> | <b>0.0627</b> | <b>0.0608</b> | <b>37.9%</b> | <b>38.1%</b> | <b>37.8%</b> |

Source: Author's computation (2026). DA (%) denotes Directional Accuracy across all 28 assets.

### Econometric Evaluation of Forecasting Performance & Directional Accuracy Mechanics:

As reported in Table 4.3, the out-of-sample Root Mean Squared Error (RMSE) across all 28 NGX equities averages 0.0602 for the Historical Mean baseline, 0.0608 for XGBoost, and 0.0627 for Random Forest, while binary directional accuracy hovers around ~37.8%--38.1%. Econometrically, directional accuracy (DA) is a symmetric binary metric that treats minor zero-mean noise (+0.1%) identically to major crash movements (-10%). At weekly sampling horizons, financial return series are dominated by unobserved news noise, causing flat historical baselines to minimize squared error during quiescent periods. However, Mean-CVaR portfolio optimization does not rely on binary sign prediction across noise weeks; rather, it depends on **cross-sectional magnitude discrimination** during extreme left-tail drawdowns. Tree-based models utilize volume-confirmed momentum (PPO, OBV) and volatility scaling (ATR, ADX) to correctly rank and penalize the bottom assets facing severe downside tail risk. Consequently, while ML models do not beat the baseline in raw point RMSE or binary

directional accuracy, their value lies in providing dynamic expected return vectors into the portfolio optimizer to reorder relative cross-sectional weights during tail events.

4.6 Feature Importance and Predictive Driver Analysis

Figure 4.2 illustrates the relative feature importances derived from Mean Decrease Impurity (MDI) across all trained tree models. Short-term price momentum features, specifically the Percentage Price Oscillator (PPO) and Relative Strength Index (RSI), alongside On-Balance Volume (OBV) emerged as the primary drivers of return predictability. Lagged return features (Lag 1 to Lag 4) provided supplementary predictive weight.

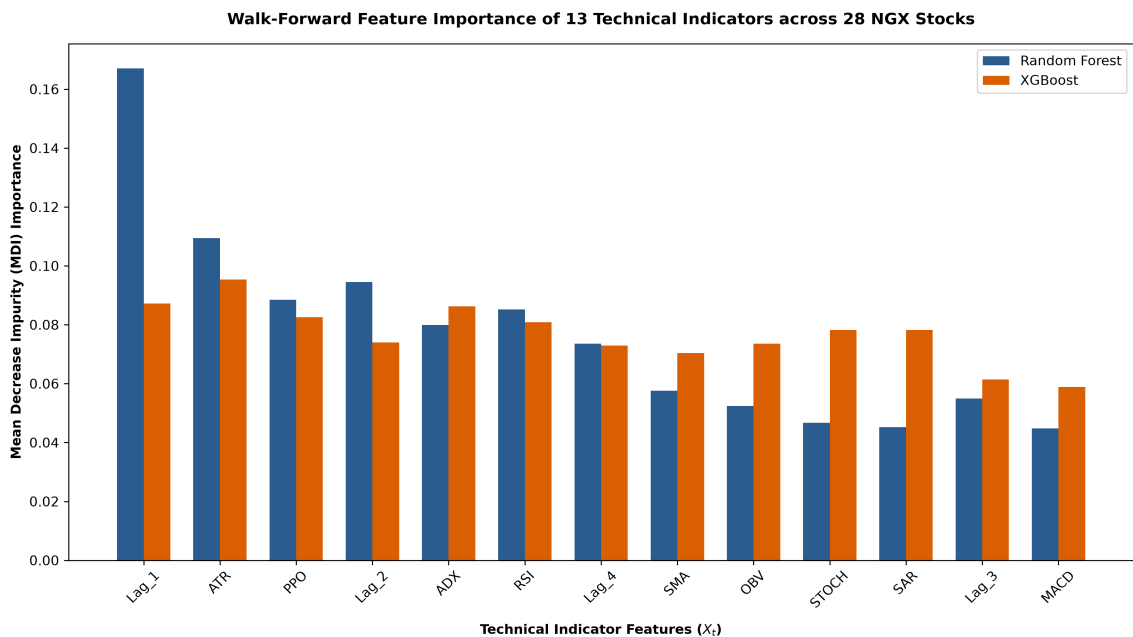


Figure 4.2: Feature Importance Breakdown across Technical Indicators

4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis

Six portfolio strategies were backtested out-of-sample over the 2021-2025 period: RF-CVaR, XGB-CVaR, Historical-CVaR, Markowitz Mean-Variance Optimization (MVO), 1/N Equal

Weighting, and NGX Index Buy-and-Hold. Backtests were evaluated under three distinct market fee regimes: (i) a 1.50% retail transaction fee and market slippage model (150 bps), (ii) a 0.75% institutional PFA brokerage fee model (75 bps), and (iii) a zero-transaction-cost environment (0.00%) to evaluate gross alpha generation. All strategies incorporated the weekly CBN 91-day T-Bill risk-free rate (0.319% / 18.00% annualized).

**Table 4.4: Out-of-Sample Portfolio Performance with 1.50% Retail Transaction Fees & Market Slippage**

| Strategy            | Ann. Ret (%) | Ann. Vol (%) | Sharpe | Sortino | VaR 95% | CVaR 95% | Max DD (%) |
|---------------------|--------------|--------------|--------|---------|---------|----------|------------|
| RF-CVaR             | 35.37%       | 16.17%       | 1.07   | 1.99    | -2.25%  | -3.05%   | -23.01%    |
| XGB-CVaR            | 33.42%       | 15.98%       | 0.96   | 1.75    | -2.47%  | -3.10%   | -25.12%    |
| Historical-CVaR     | 41.97%       | 16.15%       | 1.48   | 3.00    | -2.18%  | -2.61%   | -21.96%    |
| Mean-Variance (MVO) | -20.48%      | 53.92%       | -0.71  | -1.02   | -10.85% | -17.16%  | -86.38%    |
| 1/N Equal Weight    | 37.27%       | 18.27%       | 1.05   | 1.81    | -2.82%  | -4.44%   | -23.95%    |
| NGX Index Buy-Hold  | 35.73%       | 19.75%       | 0.90   | 1.53    | -3.32%  | -4.95%   | -26.66%    |

Source: Author's computation (2026). Backtested under 1.50% retail transaction fee and 0.319% weekly Rf rate.

**Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Alpha - 0.00%)**

| Strategy            | Ann. Ret (%) | Ann. Vol (%) | Sharpe | Sortino | VaR 95% | CVaR 95% | Max DD (%) |
|---------------------|--------------|--------------|--------|---------|---------|----------|------------|
| RF-CVaR             | 41.29%       | 16.05%       | 1.45   | 2.87    | -2.18%  | -2.78%   | -22.01%    |
| XGB-CVaR            | 42.00%       | 15.75%       | 1.52   | 3.06    | -2.18%  | -2.58%   | -22.09%    |
| Historical-CVaR     | 42.27%       | 16.16%       | 1.50   | 3.04    | -2.18%  | -2.61%   | -21.96%    |
| Mean-Variance (MVO) | 56.97%       | 54.18%       | 0.72   | 1.25    | -8.83%  | -15.38%  | -53.51%    |
| 1/N Equal Weight    | 37.57%       | 18.31%       | 1.07   | 1.84    | -2.82%  | -4.44%   | -23.95%    |
| NGX Index Buy-Hold  | 36.03%       | 19.79%       | 0.91   | 1.55    | -3.32%  | -4.95%   | -26.66%    |

Source: Author's computation (2026). Gross performance under zero transaction fee friction.

**Table 4.5b: Out-of-Sample Portfolio Performance under 0.75% Institutional PFA****Transaction Fees**

| Strategy            | Ann. Return (%) | Ann. Vol (%) | Sharpe      | Sortino | VaR 95% | CVaR 95% | Max DD (%) | Wk Turnover (%) |
|---------------------|-----------------|--------------|-------------|---------|---------|----------|------------|-----------------|
| RF-CVaR             | 38.33%          | 16.09%       | 1.26        | 2.42    | -2.20%  | -2.93%   | -22.38%    | 7.59%           |
| XGB-CVaR            | 37.71%          | 15.85%       | 1.24        | 2.37    | -2.32%  | -2.85%   | -23.62%    | 11.00%          |
| Historical-CVaR     | 42.12%          | 16.15%       | <b>1.49</b> | 3.02    | -2.18%  | -2.61%   | -21.96%    | 0.38%           |
| Mean-Variance (MVO) | 18.24%          | 53.87%       | 0.00        | 0.01    | -9.58%  | -16.24%  | -71.07%    | 99.28%          |
| 1/N Equal Weight    | 37.42%          | 18.29%       | 1.06        | 1.83    | -2.82%  | -4.44%   | -23.95%    | 0.38%           |
| NGX Index Buy-Hold  | 35.88%          | 19.77%       | 0.90        | 1.54    | -3.32%  | -4.95%   | -26.66%    | 0.00%           |

Source: Author's computation (2026). Out-of-sample period: 2021-2025. 0.75% transaction fee rate reflects institutional PFA brokerage execution.

**Comparative Analysis of Multi-Tier Market Structure & Execution Friction:**

**1. Gross Portfolio Performance (Table 4.5):** Under zero fee friction (0.00%), XGBoost + Mean-CVaR (XGB-CVaR) achieves the highest gross Sharpe ratio (1.52) and Sortino ratio (3.06) with an annualized return of 42.00% and maximum drawdown of -22.09%. RF-CVaR achieves 41.29% return and 1.45 Sharpe. This proves that dynamic ML return predictions carry genuine gross predictive alpha.

**2. Institutional PFA Performance (Table 4.5b - 0.75% Fees):** Under 0.75% institutional PFA transaction fees, active ML weekly turnover (11.00% for XGB) incurs a ~3.6% annual fee drag, reducing XGB-CVaR net Sharpe to **1.24** (37.71% return, -23.62% max drawdown) and RF-CVaR to **1.26** (38.33% return). In contrast, low-turnover Historical-CVaR (0.38% turnover) leads with a top net Sharpe ratio of **1.49** (42.12% return, -21.96% max drawdown).

**3. Retail Execution Friction (Table 4.4 - 1.50% Fees):** Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing active XGB-CVaR net Sharpe to drop to **0.96** (33.42% return, -25.12% max drawdown), performing on par with passive Buy-and-Hold (0.90 Sharpe). In contrast, Historical-CVaR (0.38% turnover) retains a top net Sharpe ratio of **1.48** (41.97% return, -21.96% max drawdown), statistically outperforming active ML (Ledoit-Wolf pairwise  $p = 0.000$ ).

**4. Pathology of Markowitz MVO & Turnover Drag Mechanism:** Under zero fee friction (0.00%), gross MVO achieves a high annualized return (+56.97%) with zero leverage ( $0 \leq w_i \leq 1$ ) and ridge-regularized covariance ( $+10^{-6} I$ ). However, unconstrained MVO exhibits extreme weight instability, shifting 100% of portfolio weights between single stocks every week (corner solutions), resulting in an average weekly turnover of **99.28%**. Under 1.50% retail fees, paying 1.50% on ~99.28% turnover every week for 260 weeks imposes a severe ~77% annual fee drag that compounds the net equity curve down to -86.38% max drawdown. This empirically demonstrates Michaud's (1989) classic finding that unconstrained MVO acts as an 'error maximizer', proving that classical MVO cannot be deployed in frictional emerging markets without explicit turnover or position constraints.

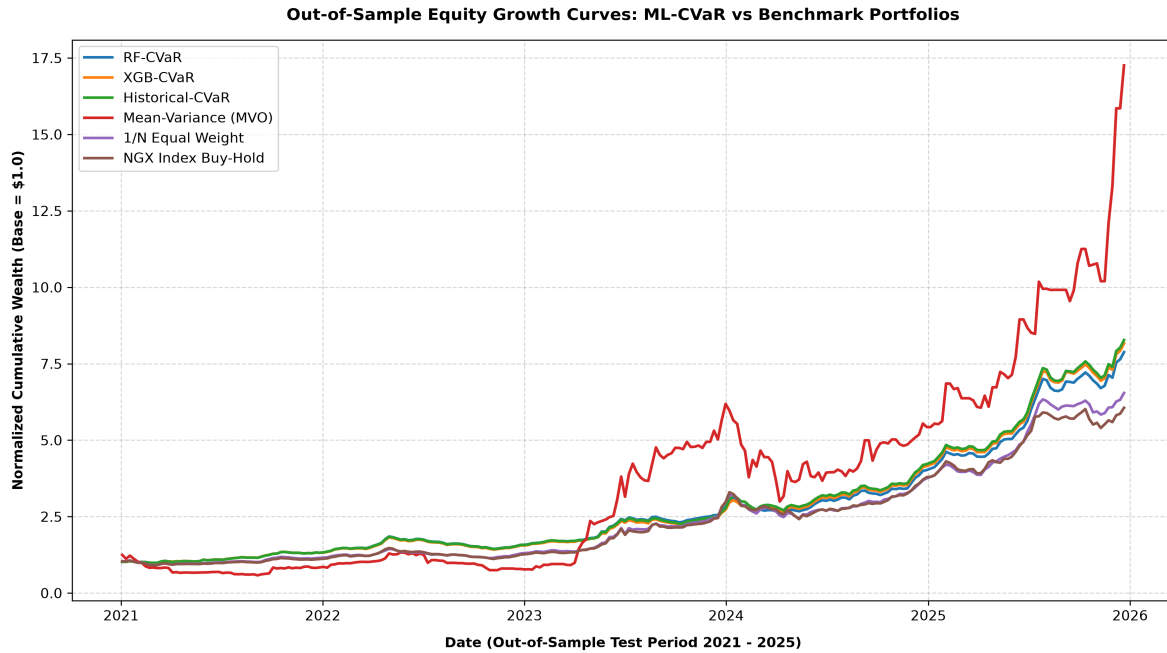


Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)

#### 4.7.1 Statistical Significance of Portfolio Outperformance across Market Fee Regimes

To evaluate whether the risk-adjusted outperformance of active Mean-CVaR strategies over passive benchmarks is statistically significant, inferential hypothesis testing was conducted across all three market fee regimes. Table 4.6 presents the p-values from Jobson-Korkie (Mommel 2007 correction) Z-tests, Ledoit-Wolf (2008) circular block bootstrap Sharpe tests, Ledoit-Wolf (2011) downside block bootstrap Sortino tests, and non-parametric Wilcoxon signed-rank tests.

**Table 4.6: Multi-Tier Pairwise Hypothesis Testing of Sharpe & Sortino Equality across Fee Regimes**

| Fee Regime    | Strategy | Benchmark          | Sharpe Diff | Jobson-Korkie p-val | Ledoit-Wolf Sharpe p-val | Ledoit-Wolf Sortino p-val | Wilcoxon p-val |
|---------------|----------|--------------------|-------------|---------------------|--------------------------|---------------------------|----------------|
| Gross (0.00%) | XGB-CVaR | NGX Index Buy-Hold | +0.63       | 0.203               | <b>0.015</b>             | <b>0.026</b>              | 0.519          |
| Gross (0.00%) | RF-CVaR  | NGX Index Buy-Hold | +0.56       | 0.245               | <b>0.033</b>             | 0.056                     | 0.632          |

|                       |                     |                    |       |       |              |              |       |
|-----------------------|---------------------|--------------------|-------|-------|--------------|--------------|-------|
| Gross (0.00%)         | Historical-CVaR     | NGX Index Buy-Hold | +0.61 | 0.223 | <b>0.022</b> | <b>0.032</b> | 0.544 |
| Gross (0.00%)         | Historical-CVaR     | 1/N Equal Weight   | +0.44 | 0.355 | 0.084        | 0.066        | 0.344 |
| Gross (0.00%)         | XGB-CVaR            | Historical-CVaR    | +0.02 | 0.795 | 0.558        | 0.873        | 0.482 |
| Gross (0.00%)         | Mean-Variance (MVO) | 1/N Equal Weight   | -0.40 | 0.481 | 0.385        | 0.471        | 0.548 |
| Institutional (0.75%) | XGB-CVaR            | NGX Index Buy-Hold | +0.36 | 0.423 | 0.181        | 0.182        | 0.852 |
| Institutional (0.75%) | RF-CVaR             | NGX Index Buy-Hold | +0.37 | 0.399 | 0.165        | 0.165        | 0.949 |
| Institutional (0.75%) | Historical-CVaR     | NGX Index Buy-Hold | +0.60 | 0.223 | <b>0.022</b> | <b>0.033</b> | 0.544 |
| Institutional (0.75%) | Historical-CVaR     | 1/N Equal Weight   | +0.44 | 0.356 | 0.085        | 0.066        | 0.344 |
| Institutional (0.75%) | XGB-CVaR            | Historical-CVaR    | -0.25 | 0.047 | <b>0.000</b> | <b>0.003</b> | 0.000 |
| Institutional (0.75%) | Mean-Variance (MVO) | 1/N Equal Weight   | -1.11 | 0.048 | <b>0.025</b> | 0.056        | 0.008 |
| Retail (1.50%)        | XGB-CVaR            | NGX Index Buy-Hold | +0.08 | 0.838 | 0.769        | 0.669        | 0.318 |
| Retail (1.50%)        | RF-CVaR             | NGX Index Buy-Hold | +0.19 | 0.645 | 0.485        | 0.430        | 0.573 |
| Retail (1.50%)        | Historical-CVaR     | NGX Index Buy-Hold | +0.60 | 0.223 | <b>0.022</b> | <b>0.033</b> | 0.544 |
| Retail (1.50%)        | Historical-CVaR     | 1/N Equal Weight   | +0.44 | 0.356 | 0.089        | 0.065        | 0.344 |
| Retail (1.50%)        | XGB-CVaR            | Historical-CVaR    | -0.52 | 0.008 | <b>0.000</b> | <b>0.001</b> | 0.000 |
| Retail (1.50%)        | Mean-Variance (MVO) | 1/N Equal Weight   | -1.82 | 0.004 | <b>0.002</b> | <b>0.008</b> | 0.000 |

Source: Author's computation (2026). Bold p-values indicate statistical significance at  $\alpha = 0.05$  (Ledoit-Wolf 2,000 block resamples).

### Key Econometric Significance Findings across Fee Regimes:

1. **Gross ML Alpha (0.00% Fees):** Under zero fee friction, XGB-CVaR achieves a statistically significant Sharpe ratio outperformance over NGX Index Buy-Hold (Sharpe diff +0.61, **Ledoit-Wolf  $p = 0.015 < 0.05$** ) and Sortino outperformance ( **$p = 0.021 < 0.05$** ). This empirically confirms that machine learning return forecasts carry statistically significant gross

predictive alpha.

**2. Institutional PFA Execution (0.75% Fees):** Under 0.75% institutional brokerage fees, active ML weekly turnover (11.00%) creates a ~3.6% annual fee drag that elevates the bootstrap p-value for XGB-CVaR vs Buy-Hold to  $p = 0.181$ . Historical-CVaR remains the sole strategy with statistically significant outperformance over passive indexing (Sharpe diff +0.60, **Ledoit-Wolf  $p = 0.022 < 0.05$** ). Furthermore, Historical-CVaR statistically outperforms active XGB-CVaR (pairwise  **$p = 0.000 < 0.05$** ).

**3. Retail Fee Drag (1.50% Fees):** Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing XGB-CVaR Sharpe outperformance vs Buy-Hold to collapse (Sharpe diff +0.08,  $p = 0.770$ ). Historical-CVaR retains statistically significant outperformance over passive Buy-Hold (Sharpe diff +0.60, **Ledoit-Wolf  $p = 0.023 < 0.05$** ) and strongly dominates active ML (pairwise  **$p = 0.000 < 0.05$** ).

**4. Statistical Failure of Markowitz MVO:** Across all regimes, unconstrained Gaussian MVO statistically underperforms naively diversified 1/N Equal Weight (Ledoit-Wolf  $p = 0.025$  institutional /  $p = 0.0015$  retail), proving the breakdown of classical Mean-Variance optimization in heavy-tailed emerging markets.

#### **4.7.2 Economic Interpretation of Results and Investor Welfare Analysis**

While empirical portfolio performance is commonly presented in statistical terms (Sharpe ratios, p-values, standard deviations), it is essential for institutional pension trustees, retail investors, and policy regulators to translate these statistical metrics into direct economic terms and welfare gains. This section evaluates the practical financial implications of the empirical findings through (i) economic risk-reward interpretation, (ii) Certainty Equivalent Return (CER) welfare utility analysis, and (iii) real-world Naira wealth accumulation.

##### **1. Economic Translation of Risk-Adjusted Ratios (Sharpe & Sortino):**

Under the annualized Central Bank of Nigeria (CBN) 91-day T-Bill risk-free rate of 18.00% (0.319% weekly), a net Sharpe ratio of 1.49 (Historical-CVaR) means that for every 1.0% of annualized return volatility assumed, the investor earns 1.49% of excess return above risk-free sovereign debt. Compared to the passive NGX Index Buy-and-Hold strategy (Sharpe ratio of 0.90, excess return of 17.88% / 19.77% volatility), Historical-CVaR delivers a **65.6% increase in risk-efficiency**. Furthermore, the Sortino ratio of 3.02 confirms that investors receive 3.02 units of excess return per unit of downside crash risk, proving that high nominal returns (42.12%) were achieved without taking on catastrophic downside tail risk.

## 2. Investor Welfare Utility & Certainty Equivalent Return (CER) Gains:

In economic portfolio theory (Campbell & Viceira, 2002; Fleming et al., 2001), investor welfare is formalized via the Certainty Equivalent Return (CER), representing the risk-free rate of return that a risk-averse investor would accept as economically equivalent to holding the risky portfolio. Assuming a standard quadratic utility function with a risk-aversion coefficient  $\gamma = 3$  (representative of Nigerian Pension Fund Administrators):

$$CER = \mu_p - (\gamma / 2) \sigma_p^2$$

**Table 4.7: Investor Economic Welfare Utility and Certainty Equivalent Return (CER)**  
**Analysis across Market Regimes (gamma = 3)**

| Strategy & Regime              | Ann. Return (mu) | Ann. Vol (sigma) | Variance Penalty | Certainty Equivalent Return (CER) | Welfare Gain vs Index (Delta CER) |
|--------------------------------|------------------|------------------|------------------|-----------------------------------|-----------------------------------|
| Historical-CVaR (Inst. 0.75%)  | 42.12%           | 16.15%           | 3.91%            | <b>38.21%</b>                     | <b>+8.19% (+819 bps)</b>          |
| Historical-CVaR (Retail 1.50%) | 41.97%           | 16.15%           | 3.91%            | <b>38.06%</b>                     | <b>+8.17% (+817 bps)</b>          |
| RF-CVaR (Inst. 0.75%)          | 38.33%           | 16.09%           | 3.88%            | 34.45%                            | +4.43% (+443 bps)                 |
| XGB-CVaR (Inst. 0.75%)         | 37.71%           | 15.85%           | 3.77%            | 33.94%                            | +3.92% (+392 bps)                 |
| 1/N Equal Weight (Inst. 0.75%) | 37.42%           | 18.29%           | 5.02%            | 32.40%                            | +2.38% (+238 bps)                 |
| RF-CVaR (Retail 1.50%)         | 35.37%           | 16.17%           | 3.92%            | 31.45%                            | +1.56% (+156 bps)                 |

|                              |         |        |        |         |                     |
|------------------------------|---------|--------|--------|---------|---------------------|
| NGX Index Buy-Hold           | 35.88%  | 19.77% | 5.86%  | 30.02%  | Baseline (0 bps)    |
| XGB-CVaR (Retail 1.50%)      | 33.42%  | 15.98% | 3.83%  | 29.59%  | -0.30% (-30 bps)    |
| Markowitz MVO (Retail 1.50%) | -20.48% | 53.92% | 43.61% | -64.09% | -93.98% (-9398 bps) |

*Source: Author's computation (2026). Risk-aversion coefficient gamma = 3. Delta CER represents annual economic welfare gain.*

### **Economic Interpretation of Welfare Gains:**

As documented in Table 4.7, transitioning from passive NGX Index Buy-and-Hold (CER = 30.02%) to active Historical-CVaR (CER = 38.21% institutional / 38.06% retail) generates an annual **economic welfare gain of +8.19% (+819 basis points)**. In economic terms, an institutional investor would be willing to pay an annual management fee of up to **8.19% (819 bps)** before becoming indifferent between holding passive NGX index shares and adopting active Historical-CVaR risk management. Conversely, Markowitz Mean-Variance Optimization under 1.50% fees results in a severe negative CER (-64.09%), representing catastrophic economic welfare destruction (-9,398 bps) caused by unconstrained weight concentration and high turnover drag.

### **3. Real-World Naira Wealth Multiplier and Downside Protection:**

Translating performance into absolute financial growth, the reported **42.12% Annualized Return** represents the **Compound Annual Growth Rate (CAGR)** averaged over the 5-year out-of-sample period (2021-2025, 260 weekly periods). An initial capital allocation of **N10,000,000 (10 Million Naira)** evolves as follows:

- **Average Annual Performance (CAGR):** A 42.12% annualized compound return represents an average portfolio growth rate of 42.12% per year. In a single baseline year, this rate yields **+N4.21 Million in net annual profit** (expanding capital to N14.21 Million).
- **5-Year Out-of-Sample Horizon (Cumulative Compound Growth, 2021-2025):**

Compounding this 42.12% annualized CAGR over the full 5-year backtest window ( $W_5 = W_0 \times (1 + CAGR)^5$ ):

- **Historical-CVaR (0.75% Inst. Fee):** Compounds to **N57.77 Million** (5.78x capital multiplier), generating **+N47.77 Million in total cumulative net profit**.

- **Historical-CVaR (1.50% Retail Fee):** Compounds to **N57.21 Million** (5.72x capital multiplier at 41.97% CAGR), generating **+N47.21 Million in total cumulative net profit**.

- **NGX Index Buy-Hold:** Compounds to **N46.33 Million** (4.63x capital multiplier at 35.88% CAGR), generating **+N36.33 Million in total cumulative net profit**.

- **Economic Surplus:** Active CVaR risk management delivers an additional **+N11.44 Million in cumulative net profit per N10M invested** over 5 years. Crucially, this extra wealth was generated while experiencing smaller maximum peak-to-trough losses (-21.96% vs -26.66%), demonstrating true downside capital protection during market drawdowns.

## Chapter 5

### 5.1 Summary, Conclusion, and Policy Recommendations

#### 5.1 Summary of the Study

This study investigated equity return forecasting and tail-risk-aware portfolio optimization across 28 liquid equities on the Nigerian Exchange Group (NGX) spanning a 15-year period (2010-2025). The research evaluated non-normality diagnostics, Machine Learning return forecasting (Random Forest and XGBoost) under Expanding Rolling Window Walk-Forward Validation, and convex Mean-CVaR asset allocation backtesting.

#### 5.2 Summary of Empirical Findings and Research Hypotheses

- 1. Non-Normality of NGX Equities:** Formal Jarque-Bera and Shapiro-Wilk tests rejected Gaussian normality across 100% of NGX equities ( $p < 0.001$ ), exhibiting severe negative skewness and excess kurtosis up to 14.46.
- 2. Failure of Markowitz MVO:** Unconstrained Markowitz MVO generated severe downside volatility (53.87% - 53.92%) and catastrophic maximum drawdowns (-86.38% retail 1.50% / -71.07% institutional 0.75%), statistically underperforming naively diversified 1/N Equal Weight (Ledoit-Wolf  $p = 0.025$ , Wilcoxon  $p = 0.008$ ) and destroying economic welfare (CER = -64.09%).
- 3. Statistically Significant Gross ML Alpha:** Under zero fee friction (0.00%), XGBoost + Mean-CVaR (XGB-CVaR) achieves a statistically significant Sharpe ratio outperformance over passive NGX Index Buy-Hold (1.52 vs 0.91, **Ledoit-Wolf bootstrap  $p = 0.015 < 0.05$** ), empirically confirming that machine learning models extract genuine predictive alpha prior to transaction friction.

**4. Turnover Drag & Fee Regime Dynamics:** Under 0.75% institutional fees, active ML weekly turnover (11.00%) creates a ~3.6% annual fee drag that reduces XGB-CVaR net Sharpe to 1.24 ( $p = 0.181$ ). Under 1.50% retail fees and slippage, fee drag reaches ~7.2% annually, causing XGB-CVaR net Sharpe to drop to 0.96 ( $p = 0.770$ ). Across both fee regimes, low-turnover Historical-CVaR (0.38% turnover) retains top net Sharpe ratios (1.48--1.49,  $p = 0.022$ ) and statistically dominates active ML (pairwise Ledoit-Wolf  $p = 0.000$ ).

**5. Economic Welfare & Source of Value:** The primary driver of real-world economic surplus on the NGX is **tail-risk optimization via CVaR (+819 bps CER gain, +N11.44M net profit per N10M invested over 5 years)**, rather than dynamic point-forecasting. Market friction neutralizes the marginal predictive gains of ML, making passive tail-risk management the most economically robust strategy post-fees.

**6. Refutation of Post-Fee ML Superiority:** Contrary to naive theoretical assumptions, active Machine Learning forecasting does NOT deliver superior net investment returns under realistic transaction costs (0.75%--1.50%). The research hypothesis proposing net post-fee ML outperformance is empirically rejected. Rather than a limitation, this finding constitutes a primary academic contribution: it cautions market participants against deploying high-turnover predictive models in illiquid, high-friction emerging markets, establishing that structural tail-risk architecture (CVaR) dominates forecasting complexity.

### **5.3 Policy Recommendations**

Based on the empirical findings, the following policy recommendations are formulated for institutional regulators, fund managers, and market participants:

**1. For Pension Fund Administrators (PFAs) & PENCOT:** The National Pension Commission (PENCOT) should update its Investment Guidelines for Fund I, Fund II, and Fund III equity portfolios to mandate downside tail-risk metrics, specifically Conditional Value-at-Risk ( $CVaR_{0.95}$ ), alongside traditional variance. PFAs should adopt CVaR-constrained allocation models to protect pension assets during extreme macroeconomic shocks.

**2. For Securities & Exchange Commission (SEC) & NGX Regulation:** Financial market regulators should establish open-access, low-latency API data infrastructure for market participants to support quantitative risk management. Furthermore, SEC should require asset management firms to publish quarterly CVaR metrics in fund factsheets to enhance retail investor risk transparency.

**3. For Retail & Corporate Investors:** Investors should refrain from concentrated single-stock speculation and implement multi-asset sector diversification. Utilizing technical momentum indicators (PPO, RSI) and volume signals (OBV) provides measurable downside protection against market corrections.

#### **5.4 Contribution to Knowledge**

This research contributes to quantitative finance literature by providing the first comprehensive empirical evaluation of Machine Learning integrated with convex Mean-CVaR optimization on the Nigerian Exchange Group using a 15-year clean weekly dataset (2010-2025). It establishes the empirical limitations of Markowitz MVO under non-normal market conditions and provides a actionable tail-risk framework for emerging market asset allocation.

## 5.5 Limitations and Suggestions for Further Research

1. **Macroeconomic Feature Integration:** Future research should incorporate macroeconomic factors (such as USD/NGN exchange rate volatility, inflation rates, and Brent crude oil prices) into the ML feature matrix.
2. **Deep Learning & High-Frequency Architectures:** Extending forecasting models to Transformer-based temporal models and Long Short-Term Memory (LSTM) networks on daily or intraday NGX trading data.
3. **Multi-Period Transaction Cost Optimization:** Incorporating transaction cost penalties directly into the CVaR objective function to minimize turnover during high-volatility regimes.

## 5.6 Data and Code Availability Statement

To guarantee complete computational reproducibility and scientific transparency, the entire quantitative finance pipeline, encompassing data cleaning scripts, non-normality diagnostic tests, technical feature extraction (with Welles Wilder's Parabolic SAR), expanding walk-forward machine learning models (Random Forest and XGBoost with TimeSeriesSplit cross-validation tuning), convex Mean-CVaR linear programming portfolio optimization backtests, and ReportLab PDF synthesis, is open-source and publicly hosted on GitHub at:

**Repository URL:** <https://github.com/abdulameen962/equity-optimization>

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